

Final Project

STUDENT PERFORMANCE ANALYSIS REPORT

Agenda

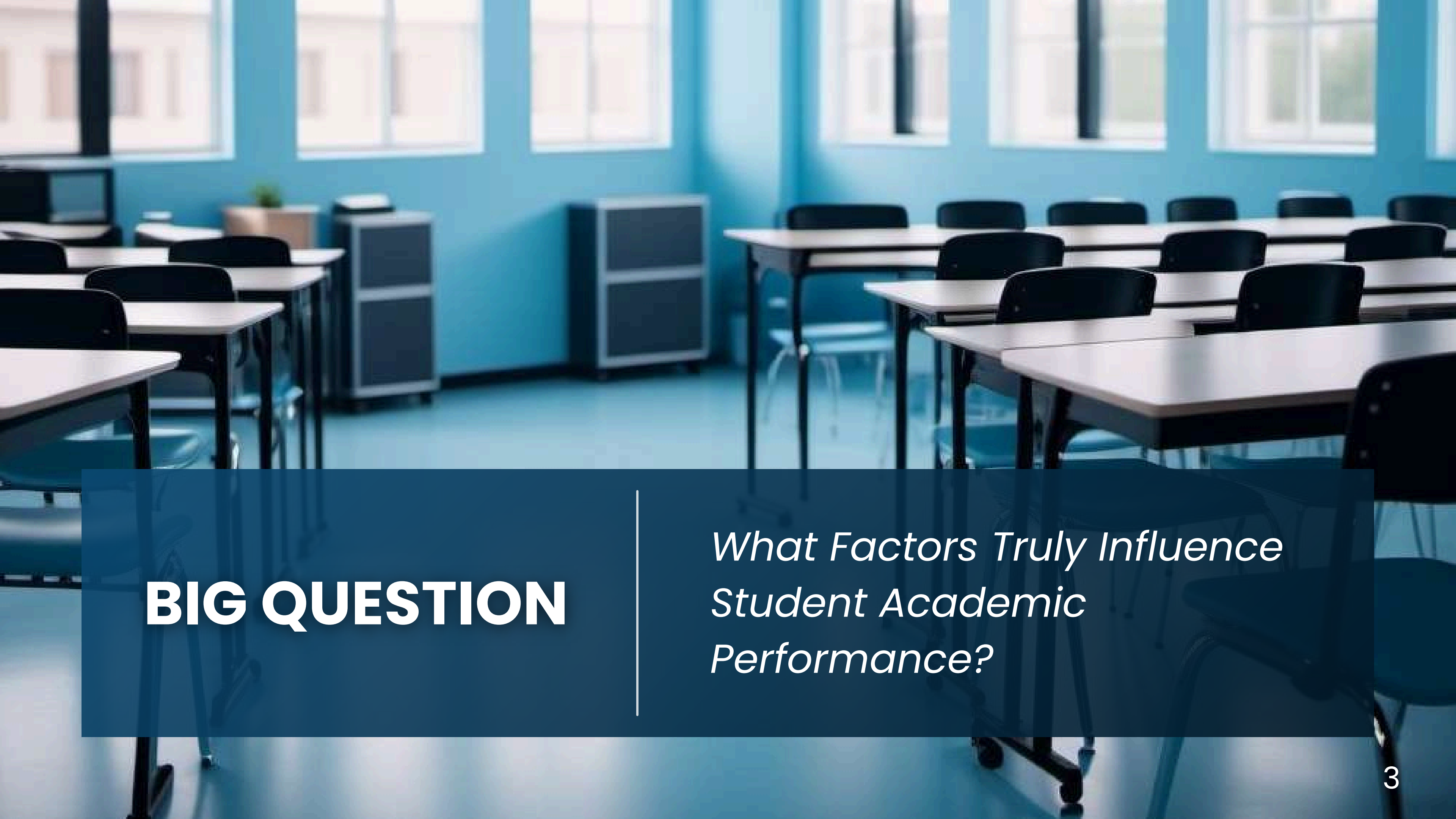
01 Introduction

02 Exploratory Data

03 Insight Data

04 Conclusion

05 Recommendations



BIG QUESTION

*What Factors Truly Influence
Student Academic
Performance?*

BIG QUESTION



How do study habits and learning motivation affect student academic performance?



How does family background shape student academic outcomes?



How do social activities and teenage lifestyle influence student performance?

1. Introduction

Dataset Background



Dataset:

- Portuguese Student Performance Dataset
- Source: UCI Machine Learning Repository



Includes:

- Student learning behavior
- Family background
- Social activities
- Lifestyle habits
- Alcohol consumption
- Academic performance

Dataset Background



Data size:

- Portuguese: 649 rows x 33 columns.
- Mathematics: 395 rows x 33 columns.



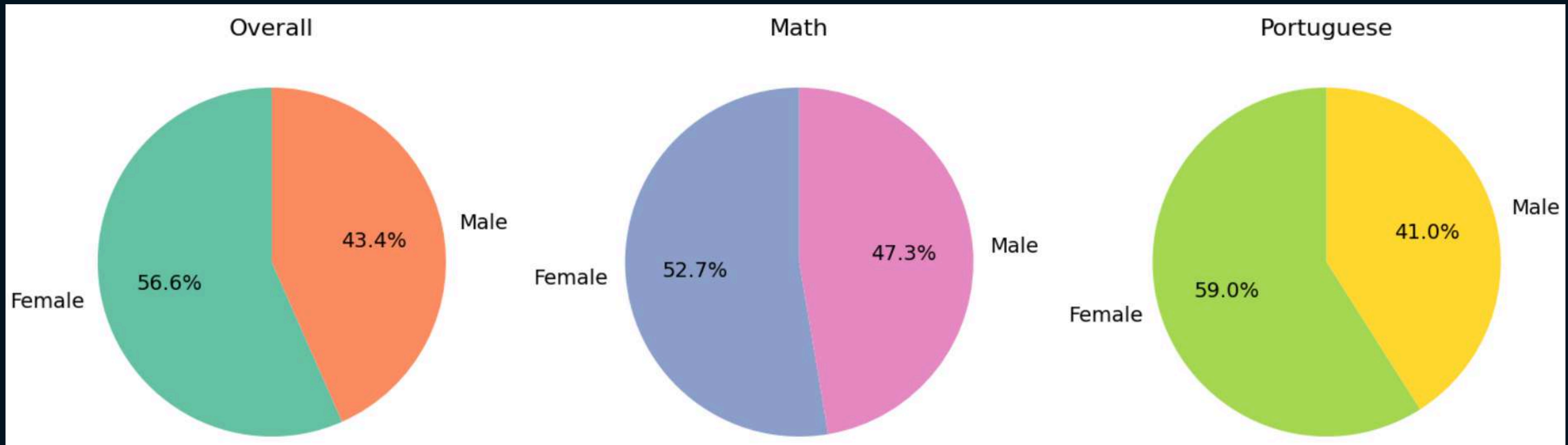
Details:

- No duplicate records.
- No missing values.

<https://github.com/Thaottm04/a/blob/55bf0168cce7a5da6ebb549d96900c814658846/final-project-data-analyst.ipynb>

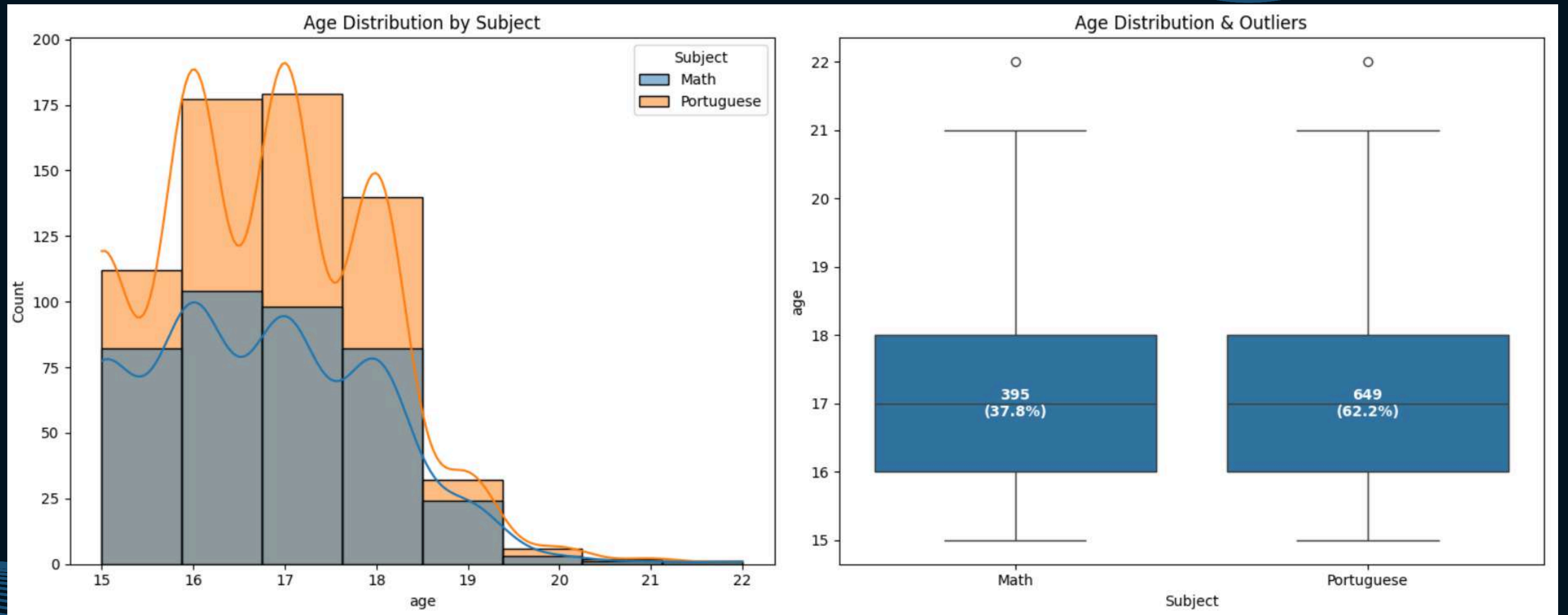
2. Exploratory Data

GENDER DISTRIBUTION



Female students make up the majority overall, accounting for nearly 60% of all students. This difference becomes even more noticeable in Portuguese classes, where female participation is significantly higher, while Mathematics classes display a more balanced ratio between male and female students.

AGE DISTRIBUTION



The age distribution shows that most students are between 15 and 18 years old. Some students are aged 20–22, suggesting delayed educational journeys or personal challenges.

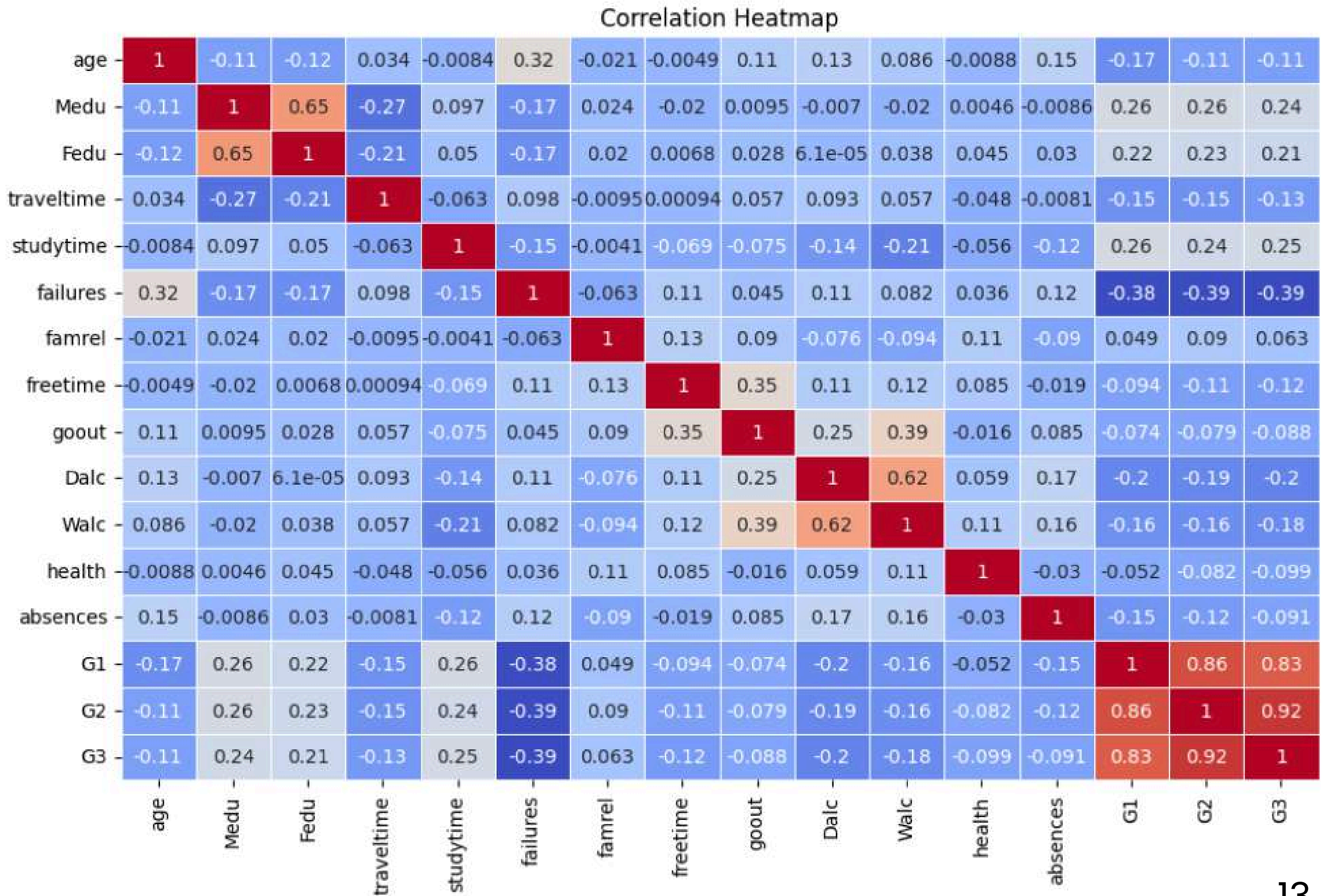
3. Insight Data



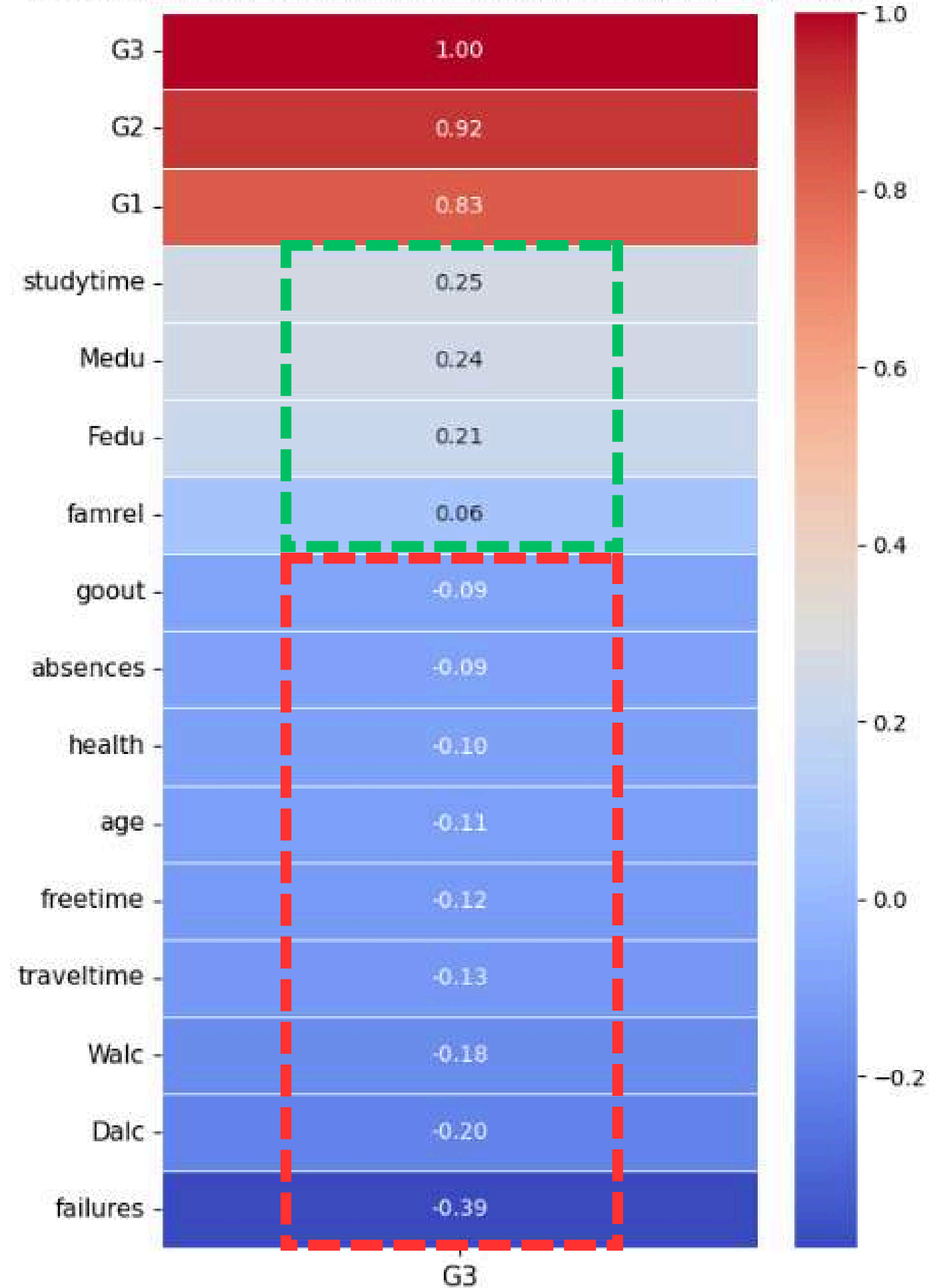
Why Focus on Portuguese?

- Larger sample size (649 students)
- Language-related subjects are more sensitive to:
 - Emotional factors
 - Family environment
 - Social life
 - Lifestyle habits

What Factors Correlate With Final Grades?



Correlation Between G3 and Other Factors



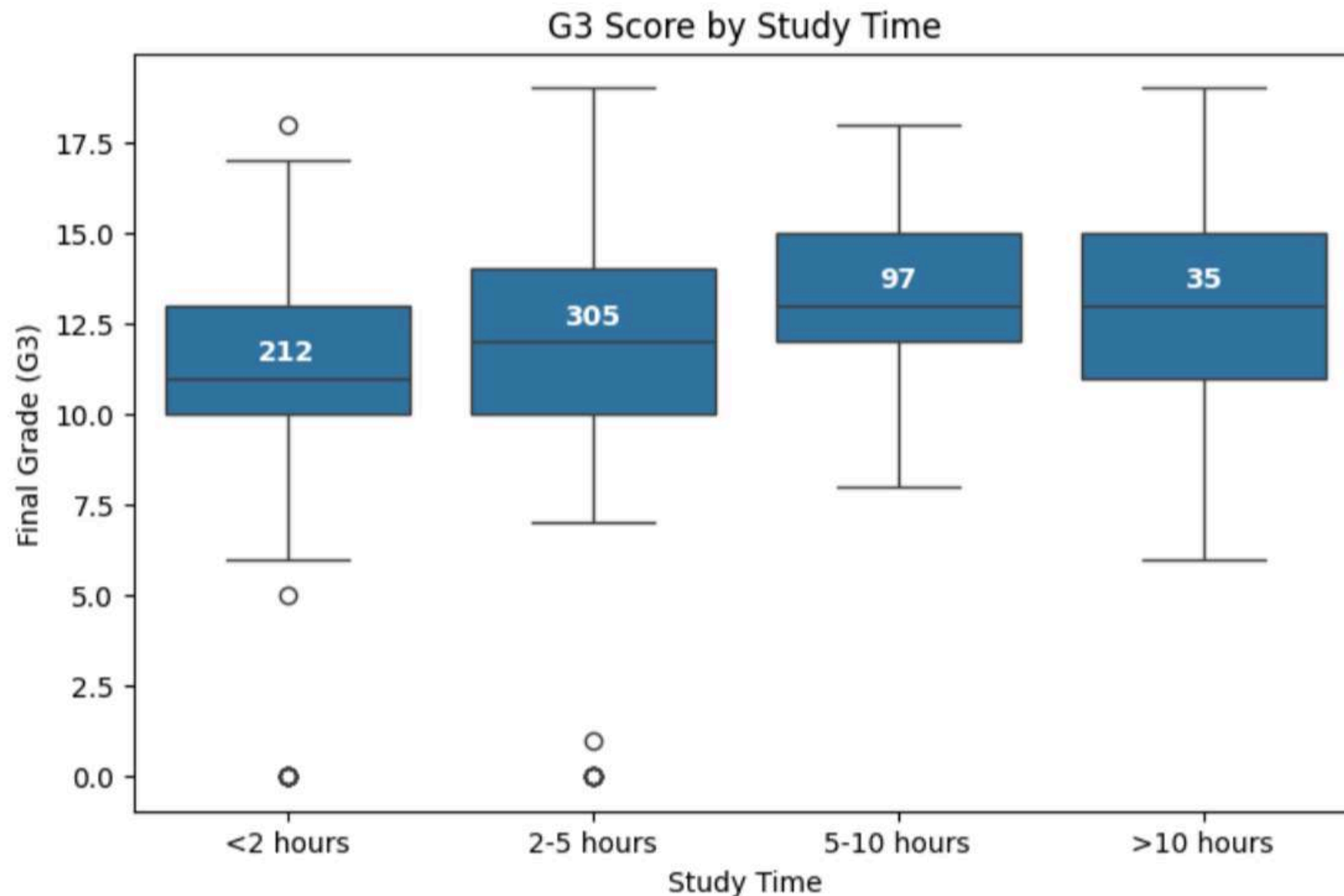
Positive Correlations

- studytime (0.25)
- Medu (0.24)
- Fedu (0.21)

Negative Correlations

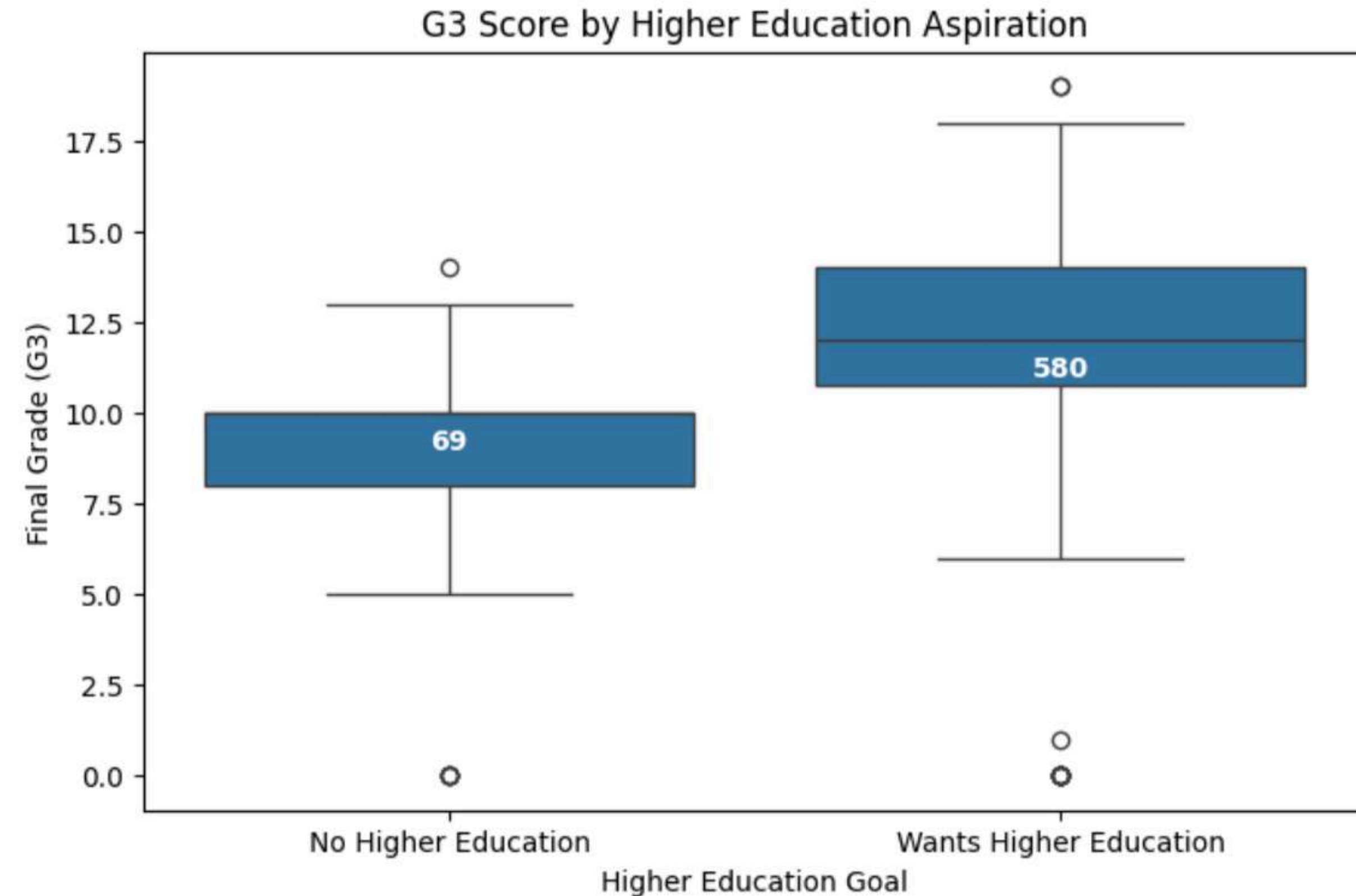
- failures (-0.39)
- Dalc (-0.20)
- Walc (-0.18)
- traveltime (-0.13)
- freetime (-0.12)
- age (-0.11)

HOW ARE STUDENTS LEARNING?



- In general, moderate studying of 2 to 10 hours weekly optimizes performance.
- Excessive study time beyond 10 hours provides no major additional benefits and risks causing student burnout.

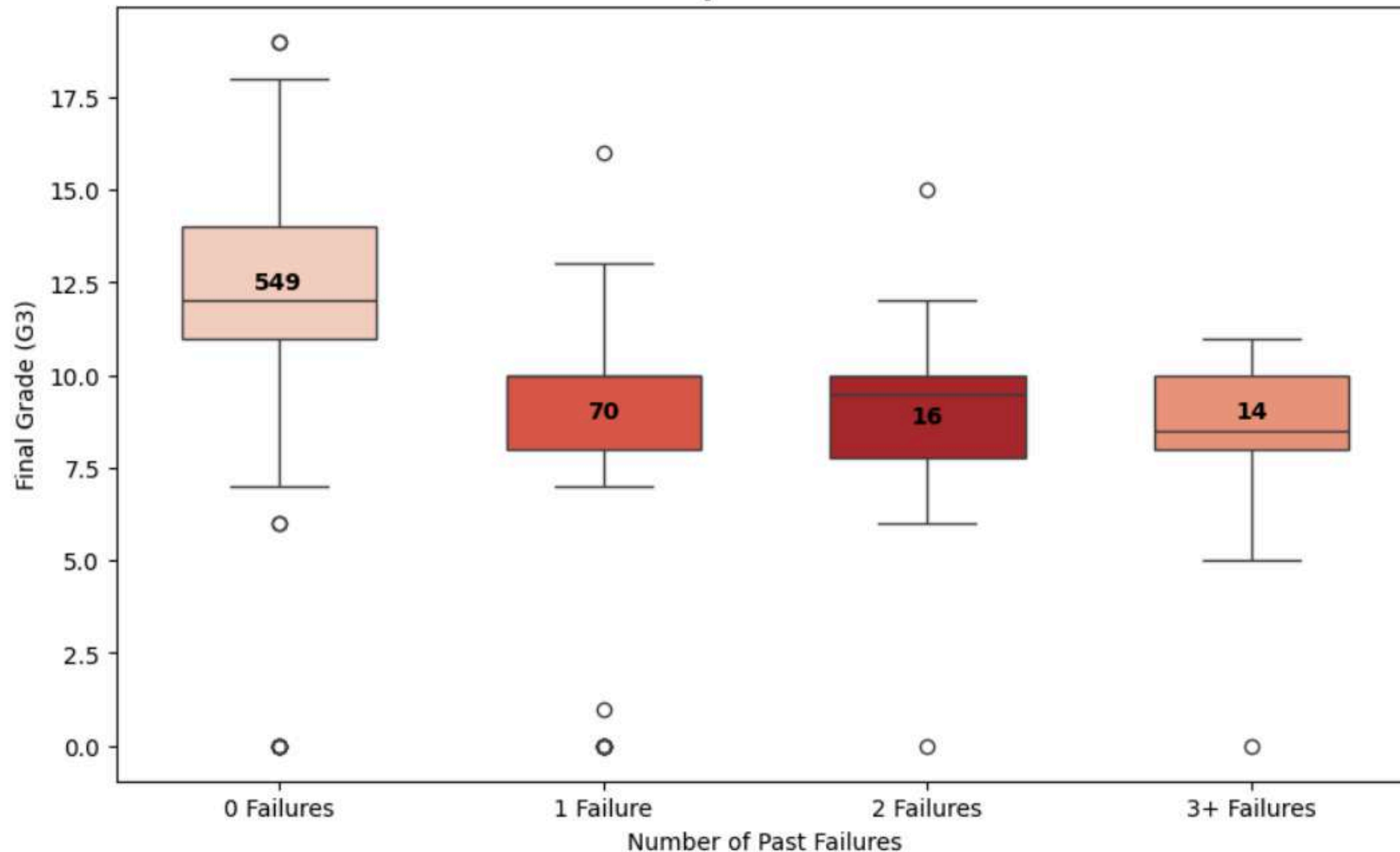
HOW ARE STUDENTS LEARNING?



- The data reveals that those aiming for university (n=580) maintain a median final score of 12, whereas the group lacking this aspiration (n=69) drops to a median score of just 10.

HOW ARE STUDENTS LEARNING?

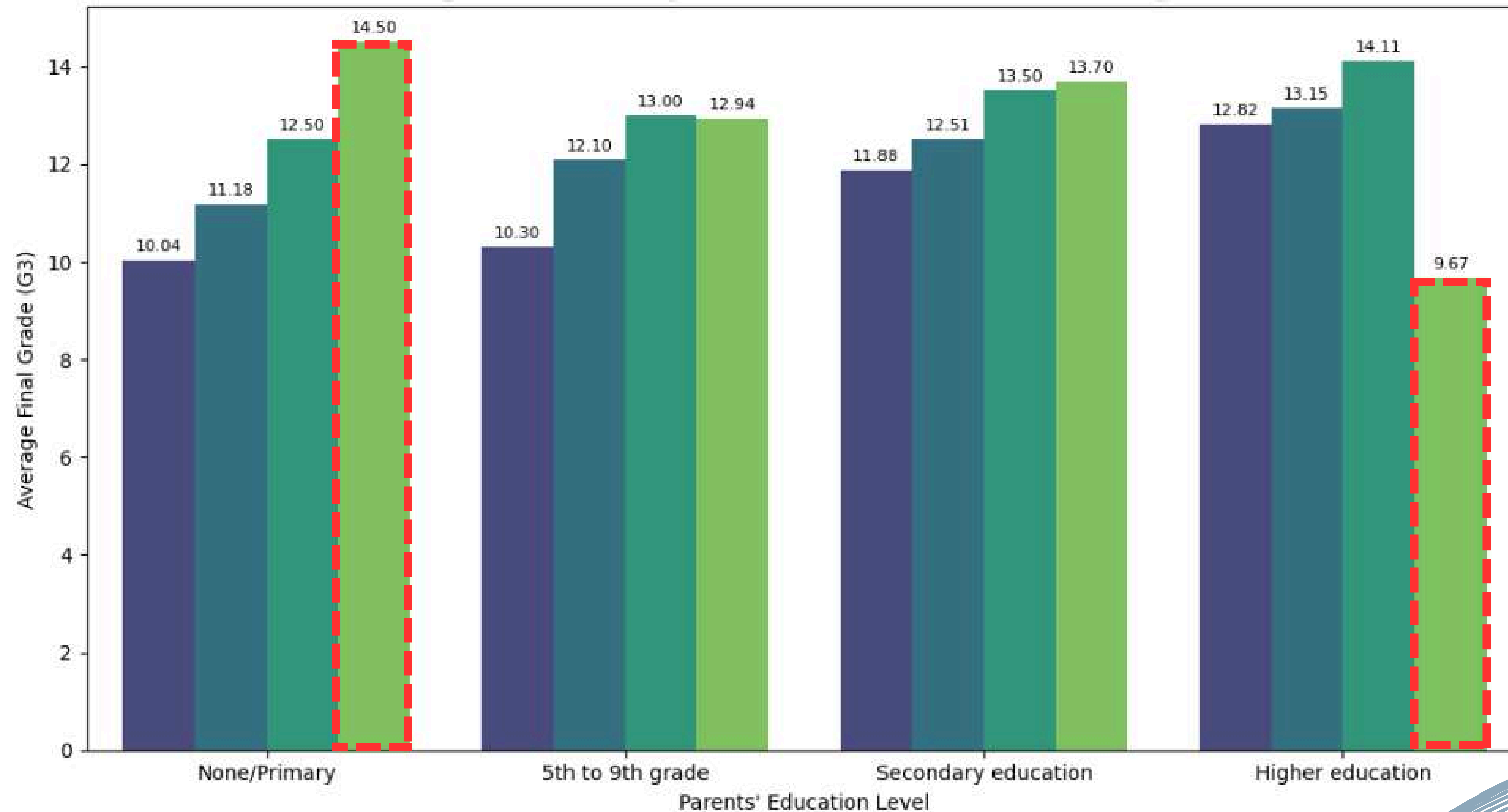
G3 Distribution by Past Academic Failures



- Students with zero past failures (n=549) achieve the highest median score of 12 and exhibit the widest grade distribution, with top marks reaching nearly 20.
- However, experiencing just one failure triggers a sharp decline, dropping the median score to 10 (n=70).

HOW ARE STUDENTS LEARNING?

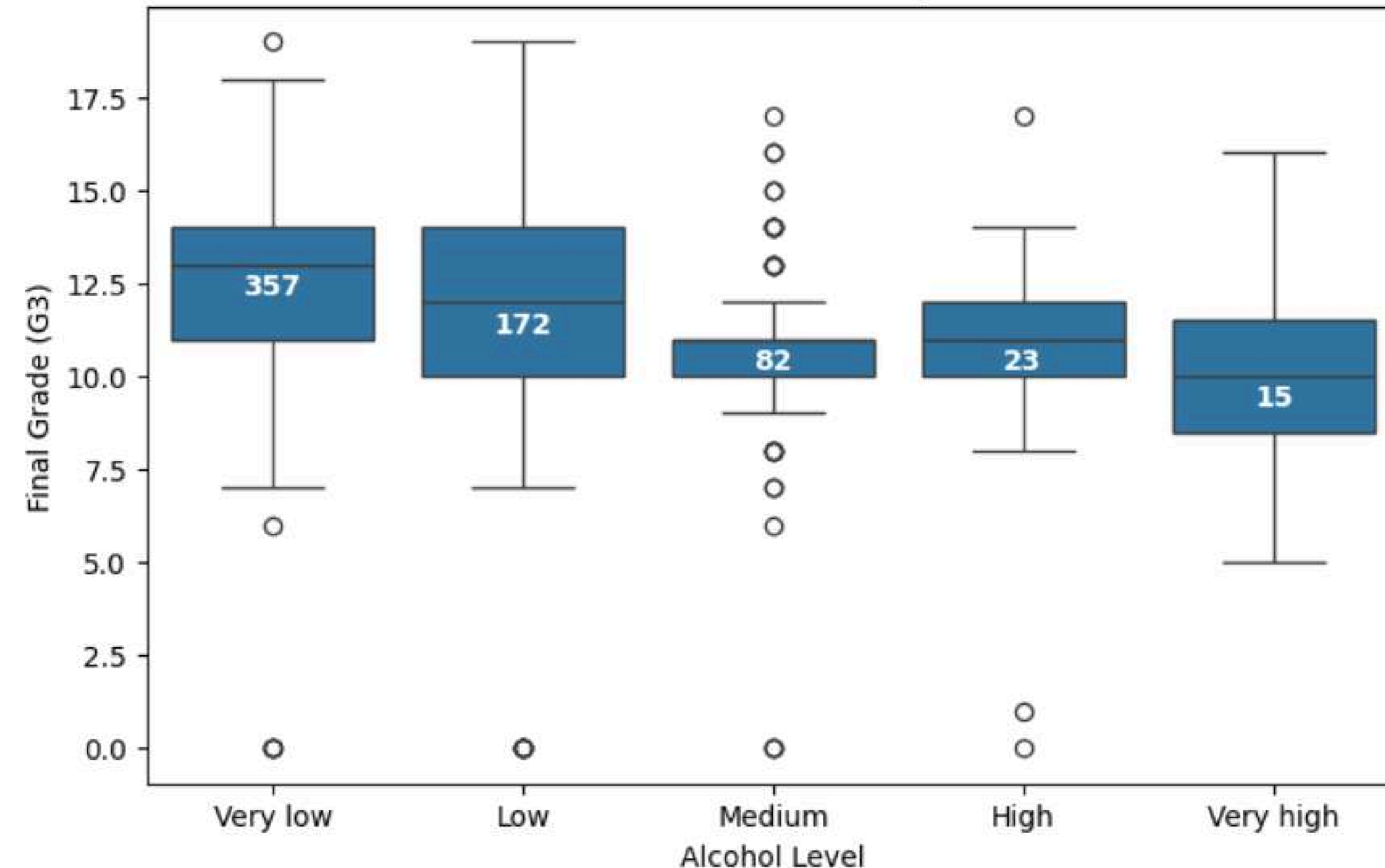
Average G3 Score by Parents' Education and Study Time



- Students from higher-education families performed best with a moderate study load of 5–10 hours/week (average grade: 14.11). However, studying >10 hours/week caused their average grade to drop sharply to only 9.67, a decrease of about 31%.

HOW DOES TEENAGE LIFE AFFECT PERFORMANCE?

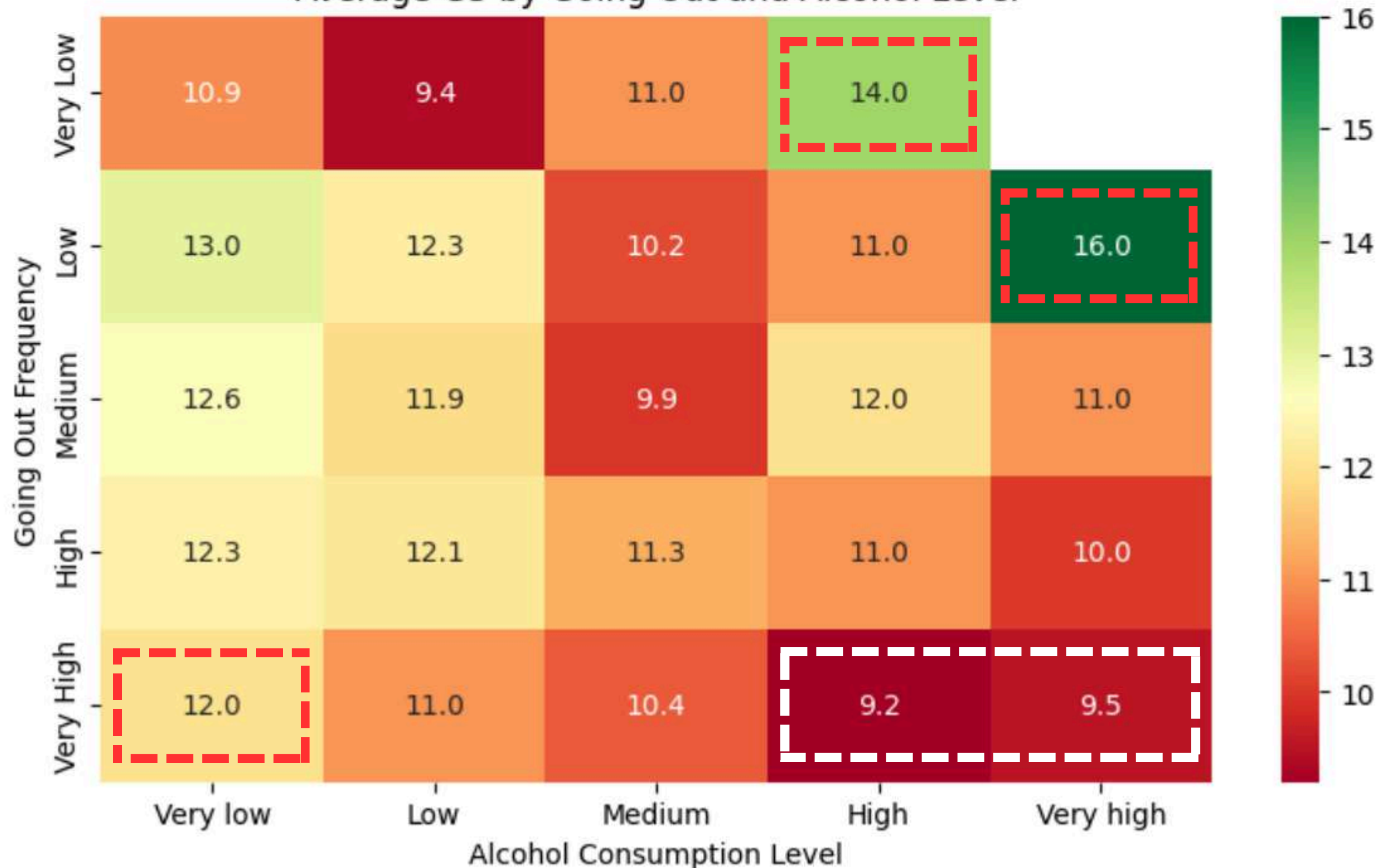
G3 vs Alcohol Consumption



- Students with very low alcohol consumption maintained a median final grade of 13, while those in the high and very high consumption groups dropped to a median grade of only 10 — a decline of approximately 23%.

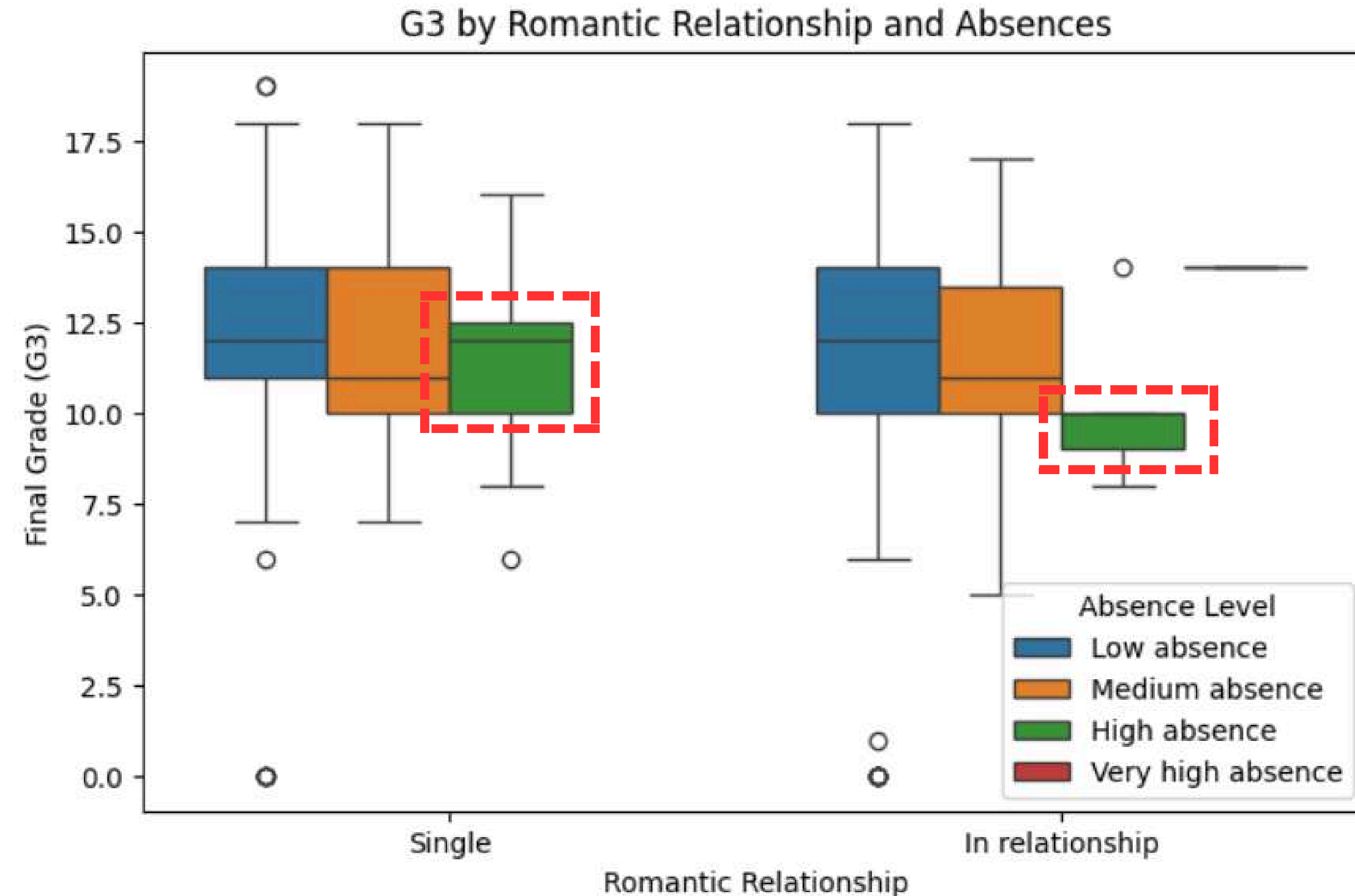
HOW DOES TEENAGE LIFE AFFECT PERFORMANCE?

Average G3 by Going Out and Alcohol Level



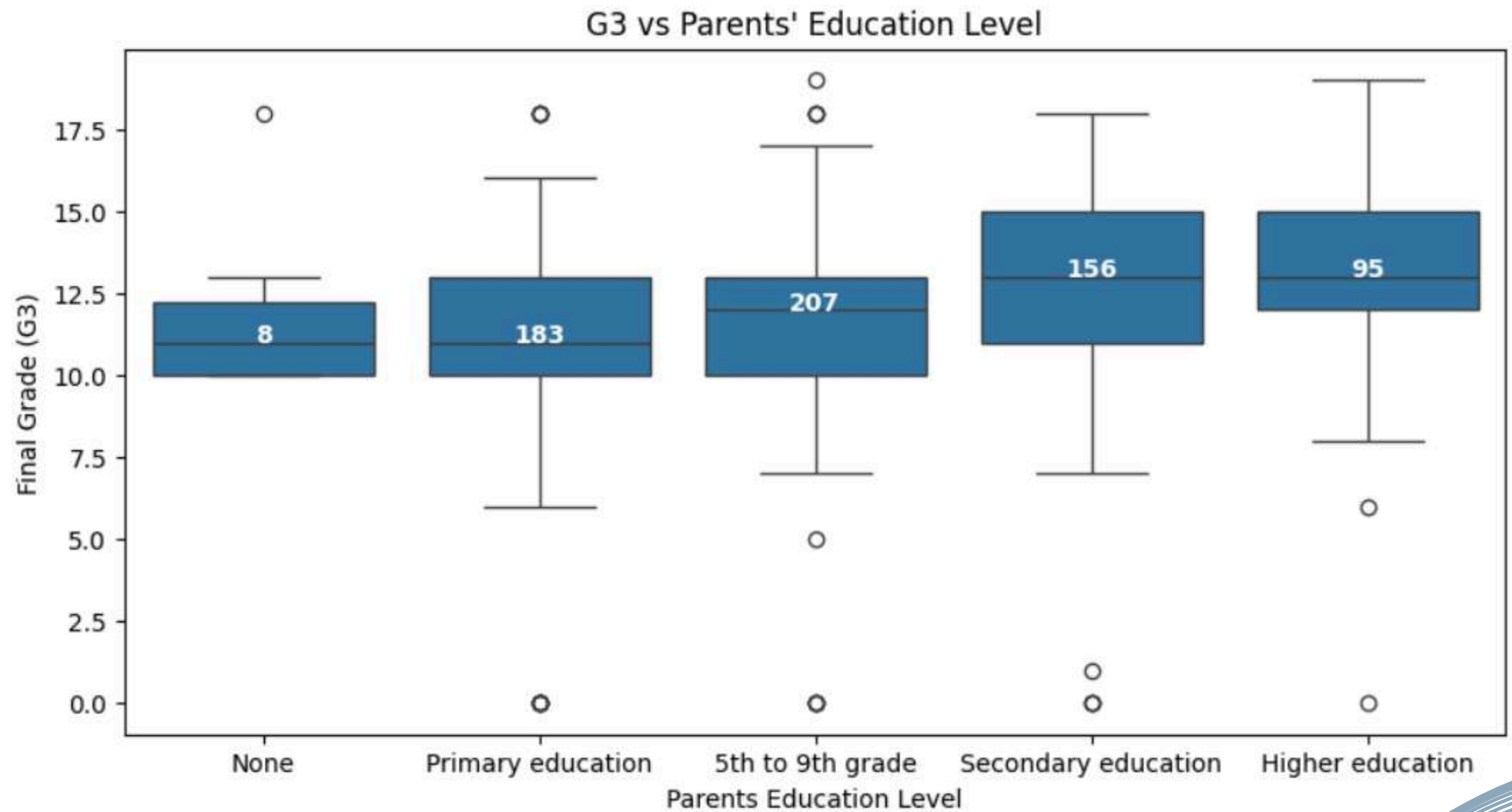
- The absolute highest average G3 score is a striking 16.0, which belongs to an unexpected demographic: students with a **"Low"** frequency of going out but **"Very high"** alcohol consumption.

HOW DOES TEENAGE LIFE AFFECT PERFORMANCE?



- While single students with **"High"** absences resiliently maintain a median score near 12, those who are both in a relationship and frequently absent suffer a severe academic decline, with their median score plummeting to 9.

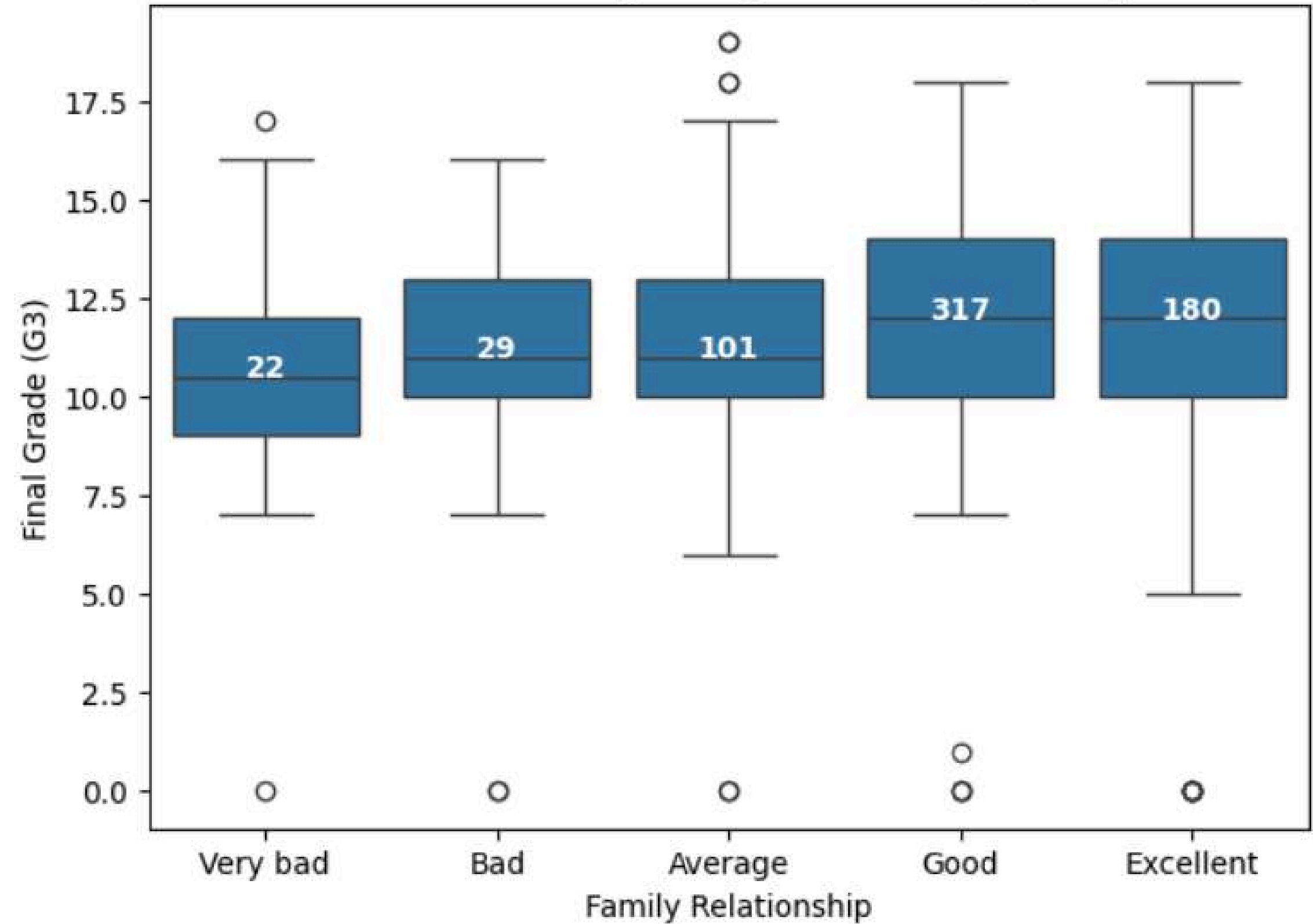
HOW DOES FAMILY INFLUENCE STUDENT PERFORMANCE?



- Students whose parents had only Primary education recorded a median final grade of 11, while students from Secondary and Higher education families reached a higher median of 13 – an improvement of about 18%.

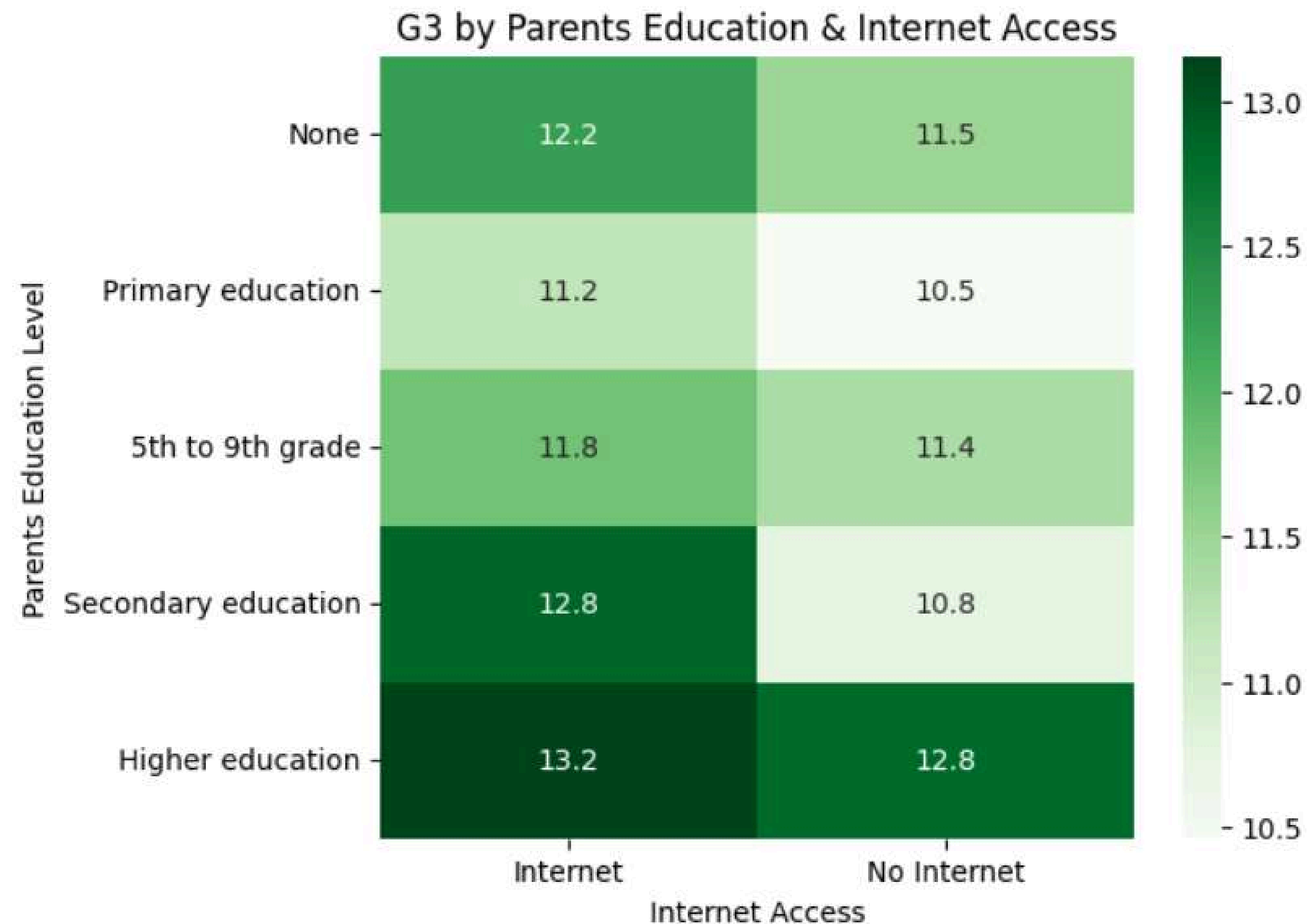
HOW DOES FAMILY INFLUENCE STUDENT PERFORMANCE?

G3 Distribution by Family Relationship Quality



- Students reporting Good and Excellent family relationships maintained a higher median final grade of 12.0, outperforming students from stressful family environments, whose medians ranged from only 10.5 to 11.0 – a gap of up to 14%.

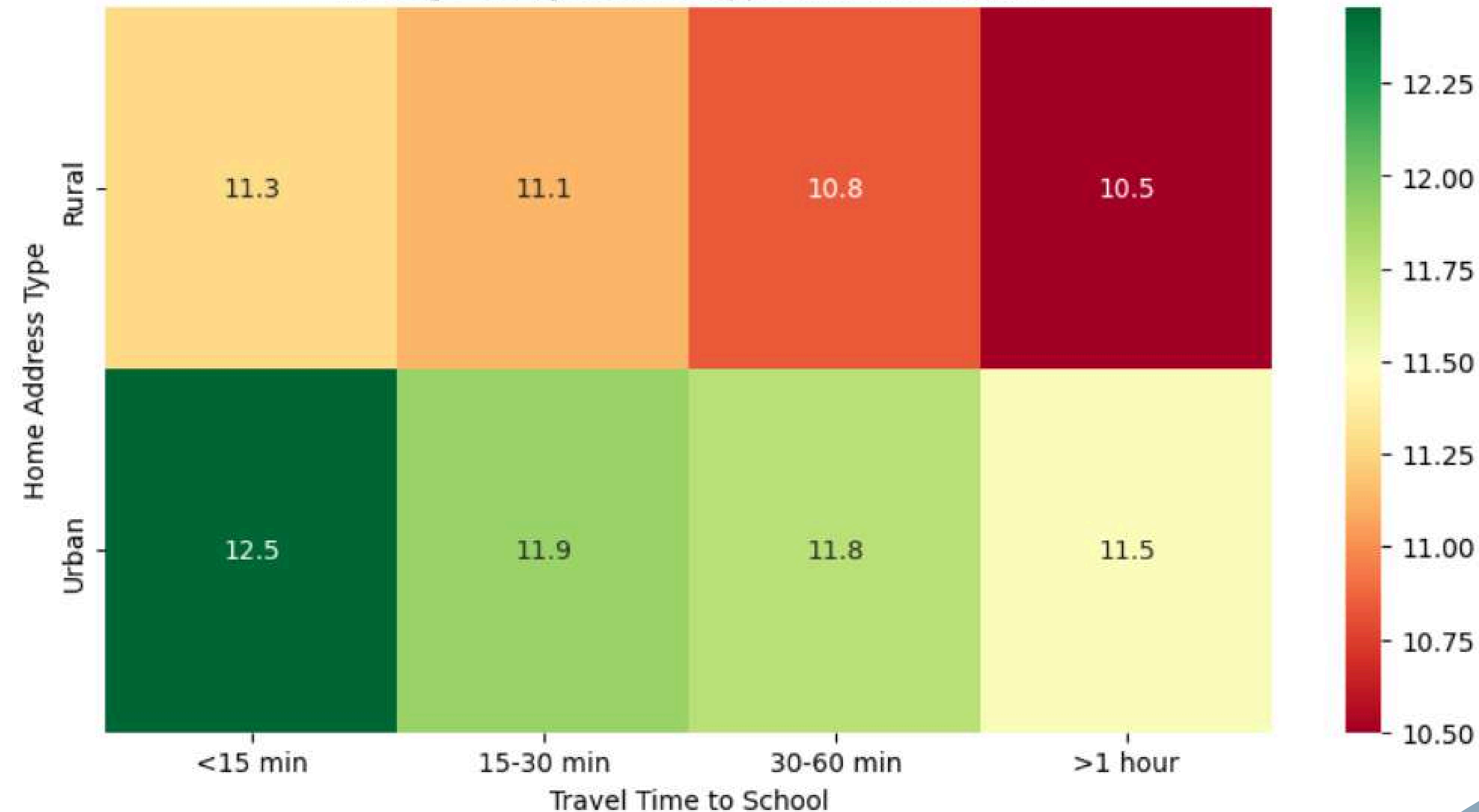
HOW DOES FAMILY INFLUENCE STUDENT PERFORMANCE?



- The largest gap appeared in the Secondary education parent group, where students with Internet access scored an average of 12.8, compared to only 10.8 for students without Internet — a difference of 2 points or approximately 18.5%.

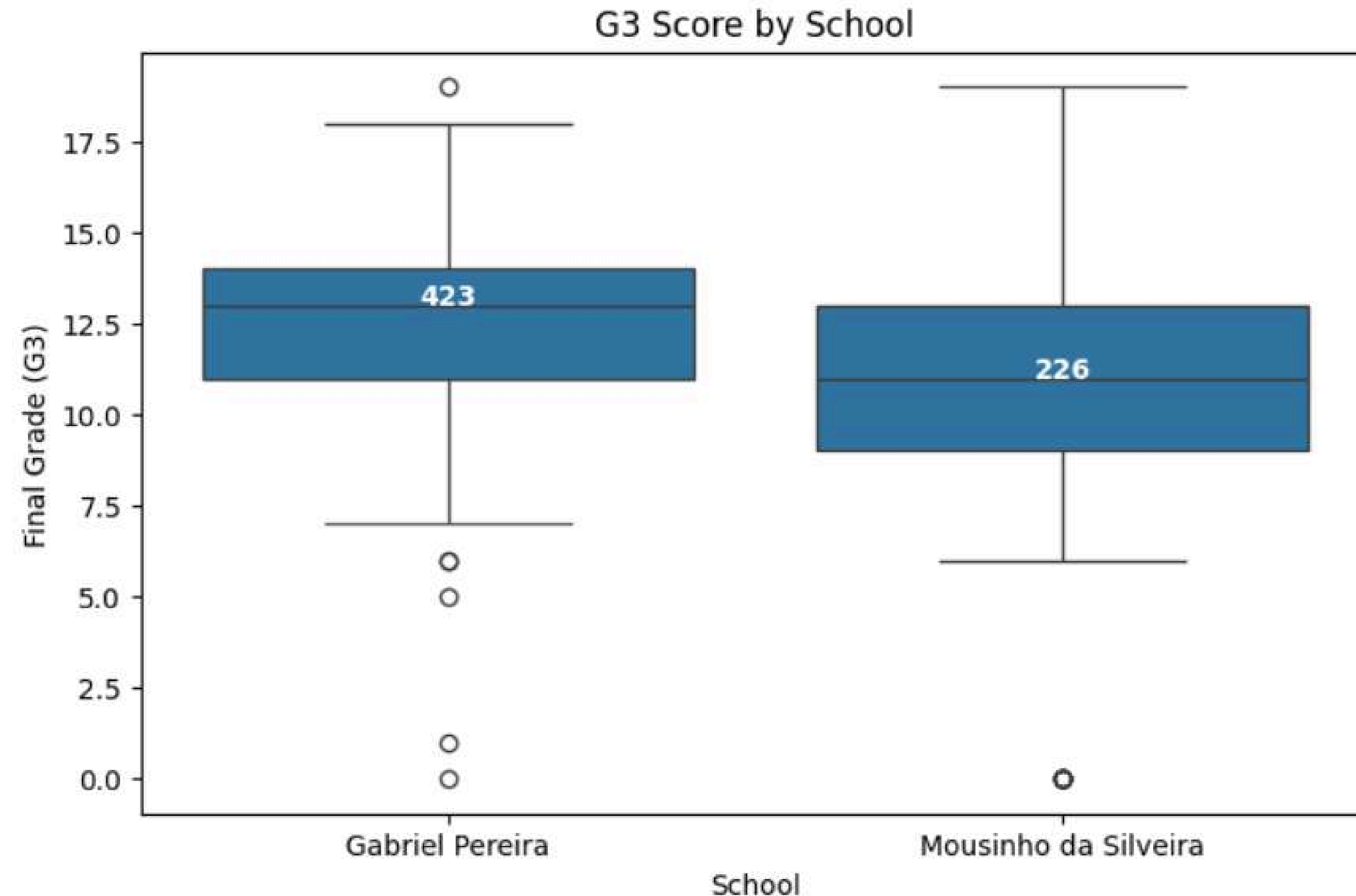
FACTORS OUTSIDE STUDENT CONTROL

Average G3 by Address Type and Travel Time



- Urban students who traveled less than 15 minutes to school achieved an average grade of 12.5, while rural students commuting for more than 1 hour averaged only 10.5 — a gap of 2 points or approximately 16%.

FACTORS OUTSIDE STUDENT CONTROL



- Students from Gabriel Pereira achieved a higher median final grade of approximately 13, while students from Mousinho da Silveira recorded a lower median of around 11 — a difference of about 18%.

4. Conclusion

BIG QUESTION–SOLVING

How do study habits and learning motivation affect student academic performance?

Students studying **5–10 hours/week** achieved the best median grade (13), while studying over 10 hours showed no further improvement. Motivation also mattered: students aiming for higher education had a median grade of 12, compared to 10 for those without future academic goals. Past failures had **the strongest negative effect**, dropping grades from 12 to 9 after just one failure.

How does family background shape student academic outcomes?

Students from Secondary/Higher education families achieved a median grade of 13, compared to 11 for students from Primary-educated families. Positive family relationships also improved performance, with students from Good/Excellent family environments maintaining a median of 12. Internet access further boosted scores, especially in Secondary-education families (**12.8 vs 10.8**).

How do social activities and teenage lifestyle influence student performance?

Lifestyle quality mattered more than socializing itself. Students with very low alcohol consumption maintained a median grade of 13, while heavy drinking reduced it to 10. Frequent socializing alone was **not harmful** (12.0 average grade), but combining it with heavy alcohol use caused grades to fall sharply to 9.2–9.5. Romantic relationships only became **harmful** when linked with high absenteeism, reducing grades to around 9.

5. Recommendations

FOR PARENTS

Recommendation 1

Parents should **pay attention** to signs of stress, loss of motivation, or burnout before academic decline becomes serious. Data shows that students with 0 past failures maintained a median grade of 12, but just 1 failure caused grades to drop sharply to around 9. In addition, students with clear higher-education goals achieved a median grade of 12, compared to only 10 among students without future academic goals.

Recommendation 2

Academic success was **strongly linked** to both **study quality** and **family stability**. Students studying 5–10 hours/week achieved the best median grade (13), while studying over 10 hours showed no additional improvement. At the same time, students from Good/Excellent family environments maintained a **stable** median grade of 12, outperforming stressful family environments (10.5–11).

Recommendation 3

Social activities themselves were **not** the main problem — unhealthy habits were. Students with very low alcohol consumption maintained a median grade of 13, while heavy alcohol use reduced it to around 10. Students who socialized frequently but avoided alcohol still averaged 12.0, whereas combining heavy socializing with alcohol dropped grades to only 9.2–9.5. This suggests parents should **focus** more on **guiding healthy lifestyles** and **setting boundaries** with harmful behaviors rather than completely restricting social interaction.



Thank You